



Supplementary Materials for
**A hydrophobic FeMn@Si catalyst increases olefins from syngas
by suppressing C1 by-products**

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This PDF file includes:

Materials and Methods
Figs. S1 to S32
Tables S1 to S6
References

Materials and Methods

Synthesis of Fe₂O₃

The nanorod-shaped Fe₂O₃ was prepared by a hard-template method. Typically, 15 g of Fe(NO₃)₃·9H₂O (AR) was dissolved in 20 mL of water. Subsequently, 4 g of carbon black (cabot vulcan xc-72), which was used as the hard template, was added in the above solution. After stirring at 50 °C for 11 h, the black product was dried in a vacuum oven at 50 °C for 6 h. Finally, the black product was calcined in static air at 550 °C for 5 h, resulting reddish brown powder. The obtained Fe₂O₃ was named as Fe.

Synthesis of Fe@Si

The core-shell Fe₂O₃@SiO₂ was prepared by the modified Stöber method. Typically, 0.5 g of the prepared Fe₂O₃ was dispersed in 300 mL of ethanol (AR) by ultrasonication. Then, 2.5 mL of tetraethoxysilane (TEOS, AR) was added. After stirring under 450 rpm for 4 h, 5 mL of ammonia (25 ~ 28%, AR) and 20 mL of water were added. The mixture was stirred under 450 rpm for another 4 h. Subsequently, the product was washed with ethanol and dried at 100 °C for 11 h. The obtained Fe₂O₃@SiO₂ was named as Fe@Si.

Synthesis of Fe@Si-c

The hydrophobic encapsulation was achieved by further organic modification. The prepared Fe@Si was preheated in a vacuum oven at 150 °C for 11 h. After cooling to 30 °C in the vacuum oven, n-hexane (AR) was added. Subsequently, chlorotrimethylsilane (TMCS, y mL per gram Fe@Si (y = 0.5, 2.0, 6.0)) was added. Then, the mixture was ultrasonic treated (40 kHz, 400W) at room temperature for 3 h. The product was washed with n-hexane and dried in a vacuum oven at 80 °C for 11 h. The obtained catalysts were named as Fe@Si-c_y. The dosage of TMCS was 2.0 mL per gram Fe@Si unless otherwise noted.

Synthesis of FeMn@Si-c

The Mn promoted Fe₂O₃ was prepared by the deposition-precipitation method. Typically, 0.5 g of the prepared Fe₂O₃ was dispersed in 300 mL of water by ultrasonication. Mn(NO₃)₂ (50%, AR) with stoichiometric amount was added. Then, the mixture was placed in an oil bath at 90 °C for 1 h. After adding 2 mL of ammonia, the mixture was aged for 11 h. The product was washed with water and dried at 100 °C for 11 h. The SiO₂-encapsulation and further organic modification were the same as for Fe₂O₃ mentioned above. The obtained catalysts were named as Fe₁Mn_z@Si-c (Mn/Fe molar ratio = z: 1, z = 0.0, 0.1, 0.3, 0.5, 1.0). The Mn/Fe ratio was 0.3 unless otherwise noted.

Catalyst characterization

Textural properties of the catalysts were measured by N₂ physisorption at -196 °C on Micromeritics ASAP 240 instrument. Prior to the measurement, the catalysts were degassed at 300 °C for 10 h under vacuum. X-ray diffraction (XRD) patterns of the powder catalysts were recorded with a PANalytical X'Pert Pro diffractometer over the 2θ range of 10° - 80°. The phases of catalysts were identified based on JCPDS standard cards. X-ray photoelectron spectroscopy (XPS) of the catalysts were recorded on a

Thermal XPS ESCALAB 250Xi Spectrometer. All peaks were corrected by setting the C 1s peak of 284.6 eV as the reference. Mössbauer spectroscopy experiments were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany).

The test process in Fig. 3D. After “reduction and carburization” or “reduction and carburization and subsequent reaction”, the reactor was cooled to room temperature under N₂ atmosphere. Then, the catalysts were collected and characterized by XRD and Mössbauer spectroscopy. Through such a test-characterization process, we can observe as much as possible the phase composition of catalysts under the actual industrial reaction conditions.

The morphology and elemental composition of the catalysts were characterized by scanning electron microscopy (SEM) on a MIRA3 TESCAN microscope equipped with an Aztec Energy X-Max 20 energy dispersive X-ray spectroscopy (EDS) detector. Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) images were obtained on a TECNAI G2 F30 with 200 kV accelerating voltage.

In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurements were accomplished in an in situ sample cell on a Thermo IS10 spectrometer. 10 mg powder catalyst was placed into the sample cell. Each measurement was performed after heating the sample cell in situ up to 400 °C under a flow of He (30 mL/min) with a ramp of 10 °C/min and keeping it at 400 °C for 2 h. The spectrum was then recorded at room temperature. Fourier transform infrared spectra (FTIR) was performed on a NICOLET 5700 FTIR Spectrometer. Prior to the measurement, the sample was mixed with potassium bromide and pressed into a wafer.

Water-droplet contact angle tests were performed on the JY-82B contact angle goniometer. An Intrinsic DVS instrument from Surface Measurement Systems was used to perform the water vapor adsorption and desorption experiments over a humidity range of 0 ~ 95%. Each humidity step was made if less than a 0.002% weight change occurred over 10 minutes, with a maximum hold time of 240 minutes.

Thermogravimetric (TG) curves were obtained on a METTLER TGA/DSC in flowing air with a heating rate of 10 °C/min. Raman spectra of the catalysts were performed on a Thermo DXR.

Catalytic tests

Fischer-Tropsch synthesis (FTS) reactions were performed in a stainless fixed-bed reactor (inner diameter = 10 mm, length = 40 cm). Typically, 0.5 g catalyst (40-60 meshes) diluted with 0.5 g quartz sand was used. Prior to reaction, the catalyst was reduced in situ at 350 °C for 10 h in a pure H₂ flow at atmospheric pressure. After reduction, the reactor was cooled to 200 °C. Then the syngas H₂/CO/N₂ (containing 5 vol% N₂ as the internal standard) was fed into the reactor. The reaction products were detected by two gas chromatographs (FULI GC 97). One gas chromatograph was equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). Porapak Q and 5A MolSieve packed column were connected to TCD, while RB-PLOT Al₂O₃ capillary column was connected to FID. The other gas chromatograph was equipped with a RB-5 capillary column and a FID. After reaction, the spent catalyst was transformed into a small glass bottle and then inflated with N₂. The bottle was put into a Teflon flask deoxygenated by Fe powder and then vacuumed and filled with N₂. The

carbon balance calculated in each test was between 96% and 98%. The CO conversion and product selectivity were calculated on a carbon-atom basis. The selectivity to oxygenates was below 2% C and has been excluded from the reported product selectivities.

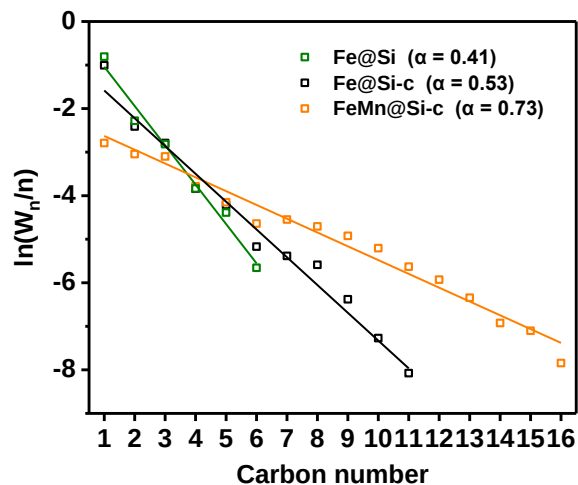


Fig. S1.

The ASF plots and relevant chain growth probability over Fe@Si, Fe@Si-c and FeMn@Si-c (corresponding to Fig. 1B). Reaction conditions: 320 °C, 2.0 MPa, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

Note: As shown in Fig. S1, Fe@Si-c exhibited a higher chain growth probability ($\alpha = 0.53$) than Fe@Si ($\alpha = 0.41$). Due to the inhibition of WGS reaction via hydrophobic treatment on Fe@Si, the H₂ amount produced by the WGS reaction decreased, suppressing the hydrogenation and chain termination reactions, thereby enhancing the chain growth probability. The addition of Mn promoter further increased the α value to 0.73 over FeMn@Si-c. Besides, the amount of CH₄ obtained over FeMn@Si-c was lower obviously than that over Fe@Si and Fe@Si-c. All of these results suggested that the hydrophobic treatment and the addition of Mn promoter effectively increased the chain growth probability and suppressed the formation of CH₄.

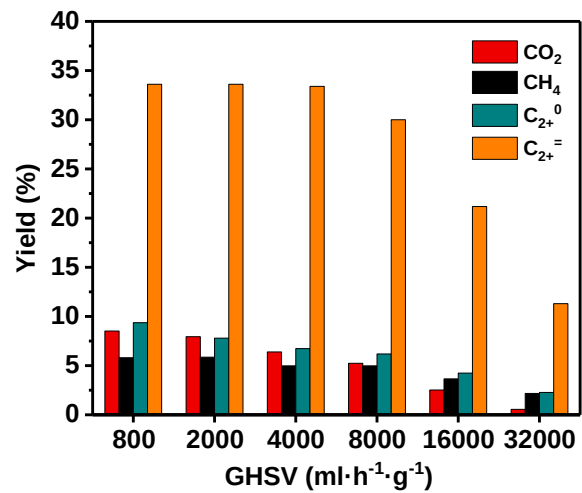


Fig. S2.

Effect of gas hourly space velocity (GHSV) on the yield of products. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

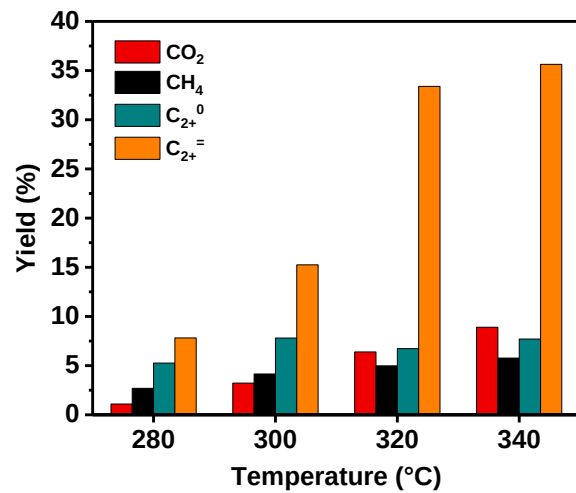


Fig. S3.

Effect of reaction temperature on the yield of products. Reaction conditions: 2.0 MPa, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

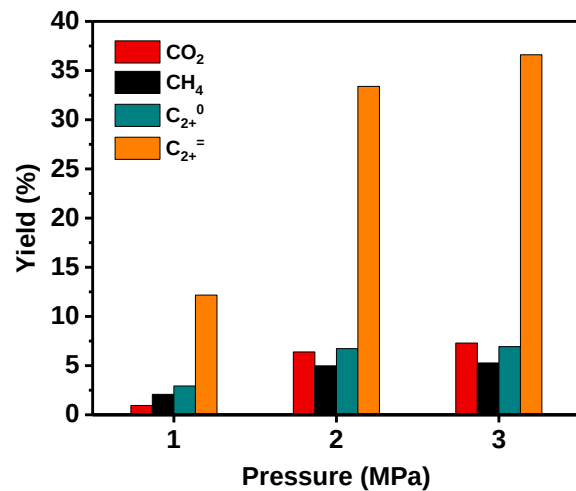


Fig. S4.

Effect of reaction pressure on the yield of products. Reaction conditions: 320 °C, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

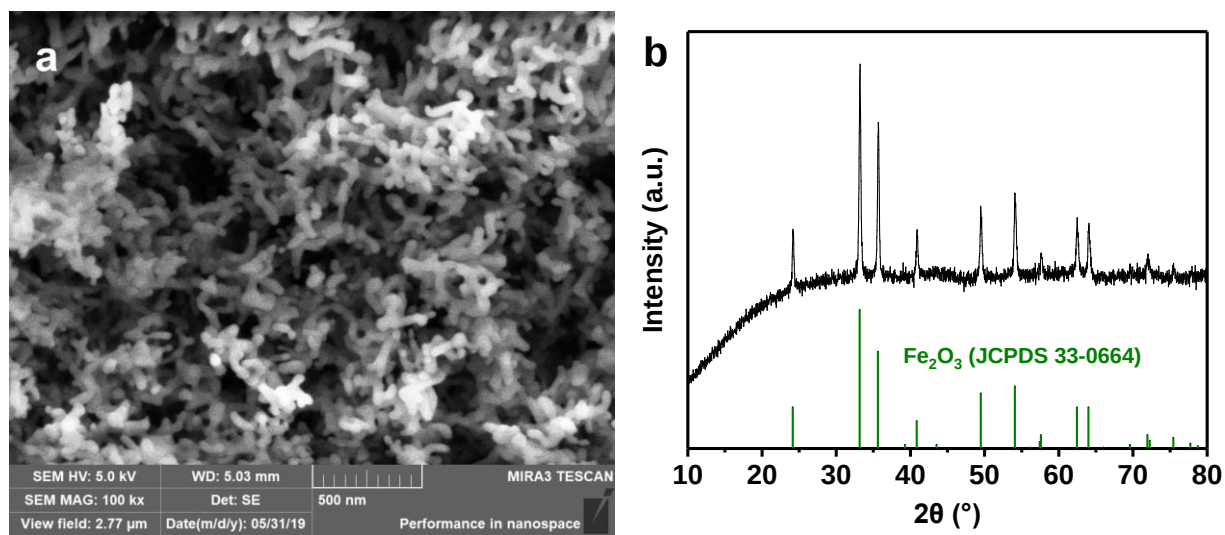


Fig. S5.

(a) SEM image and (b) XRD pattern of the Fe₂O₃ nanorods.

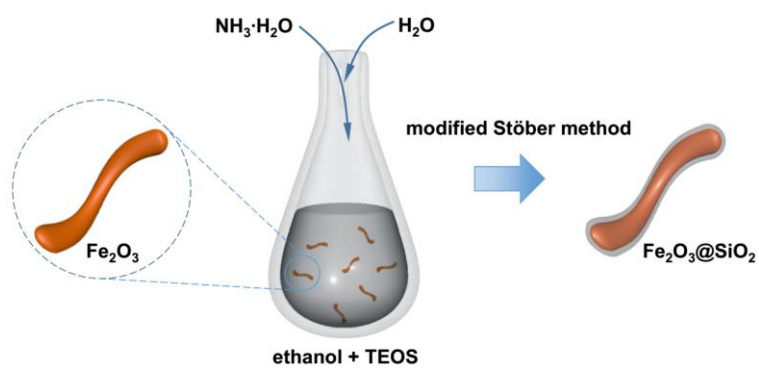


Fig. S6.
Procedures for the synthesis of core-shell $\text{Fe}_2\text{O}_3@\text{SiO}_2$.

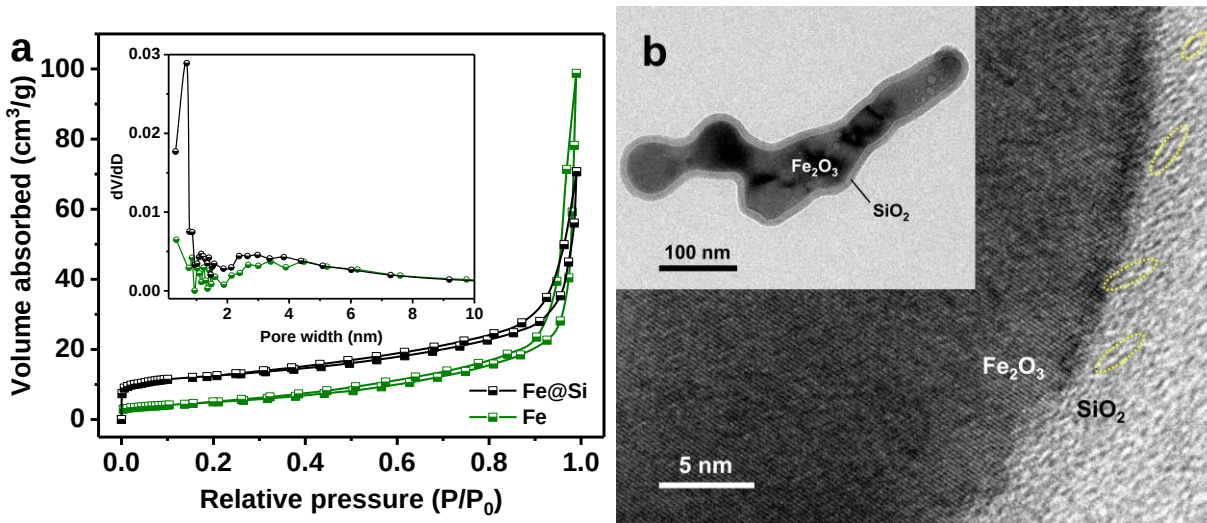


Fig. S7.

(a) N₂ adsorption-desorption isotherms and pore size distribution of Fe and Fe@Si. (b) TEM and HRTEM images of Fe@Si.

Note: As shown in Fig. S7a, after the encapsulation of SiO₂ on Fe₂O₃, more micropores with pore size ranging from 0.3 to 2 nm appeared in the pore size distribution curve, suggesting that micropores were presented in the SiO₂ shell. The HRTEM image also confirmed that many narrow micropores (marked by the golden circle in Fig. S7b) were distributed in the amorphous SiO₂ shell.

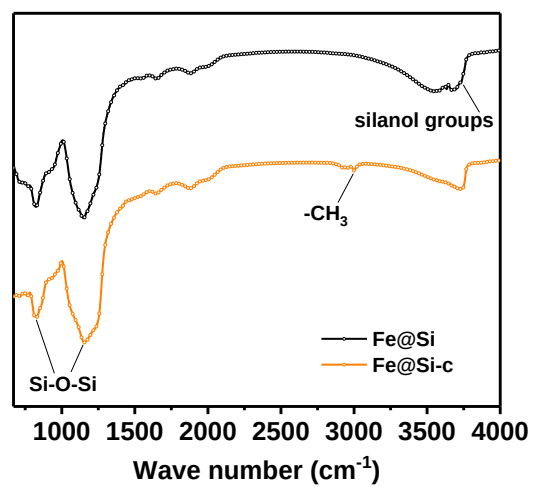


Fig. S8.
DRIFTS of Fe@Si and Fe@Si-c.

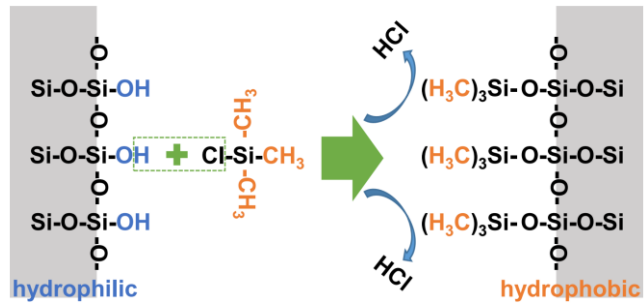


Fig. S9.

Hydrophobic modification through silanization reaction.

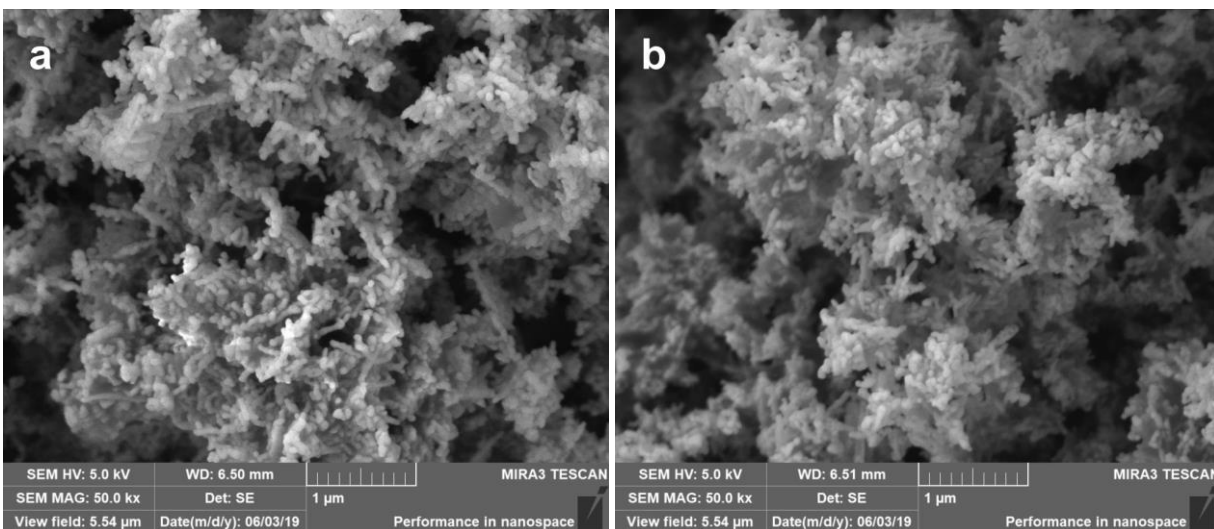


Fig. S10.

SEM images of (a) Fe@Si and (b) Fe@Si-c.

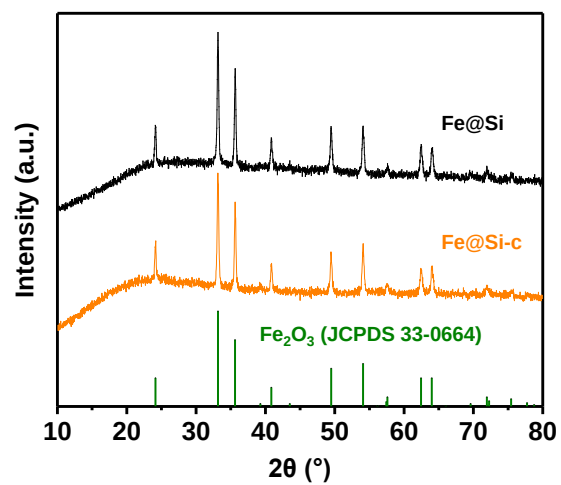


Fig. S11.
XRD patterns of Fe@Si and Fe@Si-c.

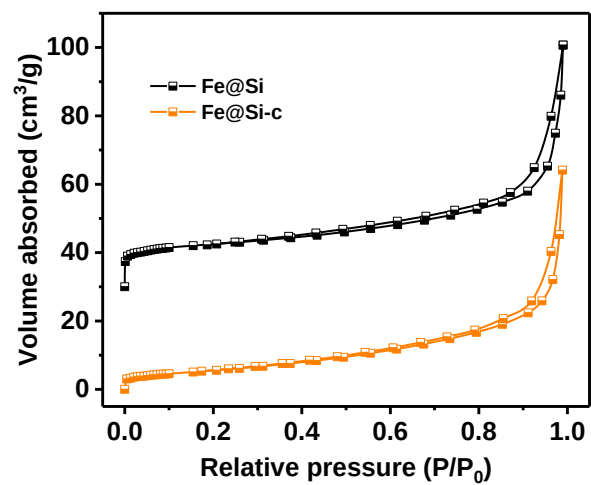


Fig. S12.

N₂ adsorption-desorption isotherms of Fe@Si and Fe@Si-c.

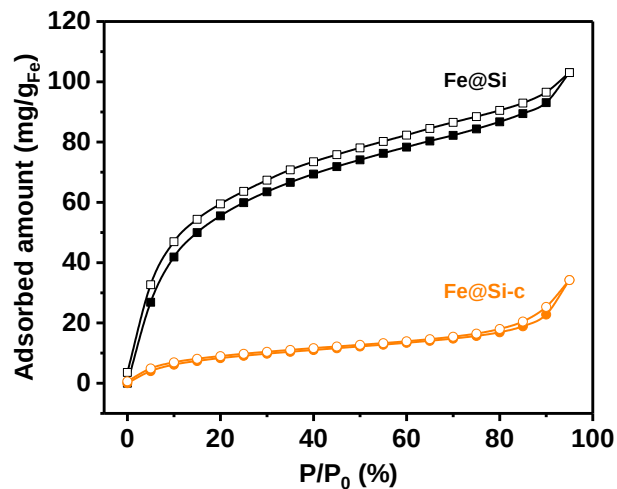


Fig. S13.

The water vapor adsorption and desorption tests on Fe@Si and Fe@Si-c. The solid and hollow markings show the adsorption and desorption data, respectively.

Note: As shown in Fig. S13, after the hydrophobic modification, the water vapor adsorption capacity of Fe@Si-c dramatically decreased compared with the hydrophilic Fe@Si. At the humidity less than 30%, the adsorbed amount of water vapor over Fe@Si was about 6.5 times that over Fe@Si-c, suggesting that the hydrophobic modification could suppress obviously the adsorption of water molecules. These results were consistent with the previous reports (35-38) that the introduced nonpolar -CH₃ groups could hinder the adsorption of polar water molecules.

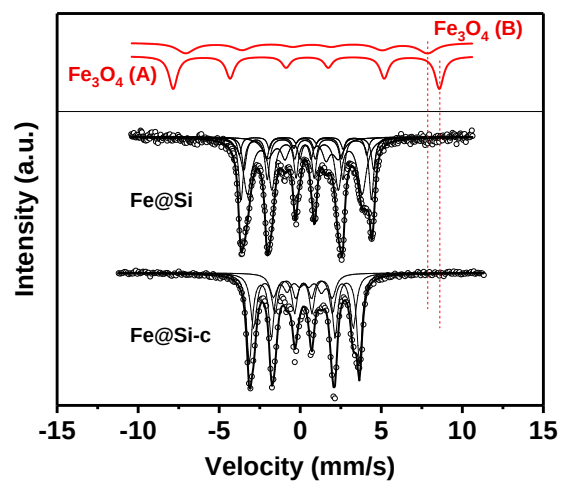


Fig. S14.

Mössbauer spectra of the catalysts after reduction and carburization (corresponding to Fig. 3D).

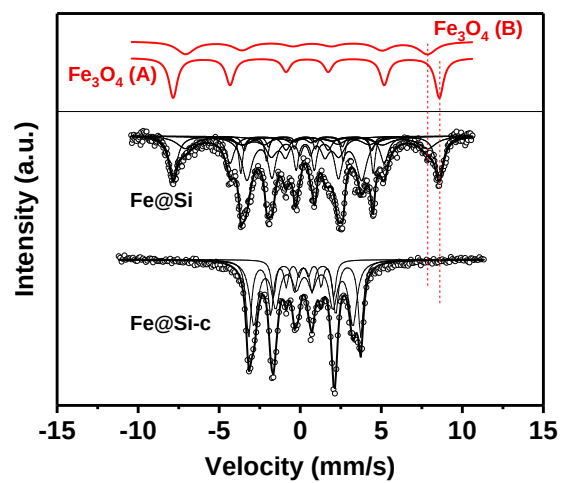


Fig. S15.

Mössbauer spectra of the catalysts after reduction and carburization and subsequent reaction (corresponding to Fig. 3D).

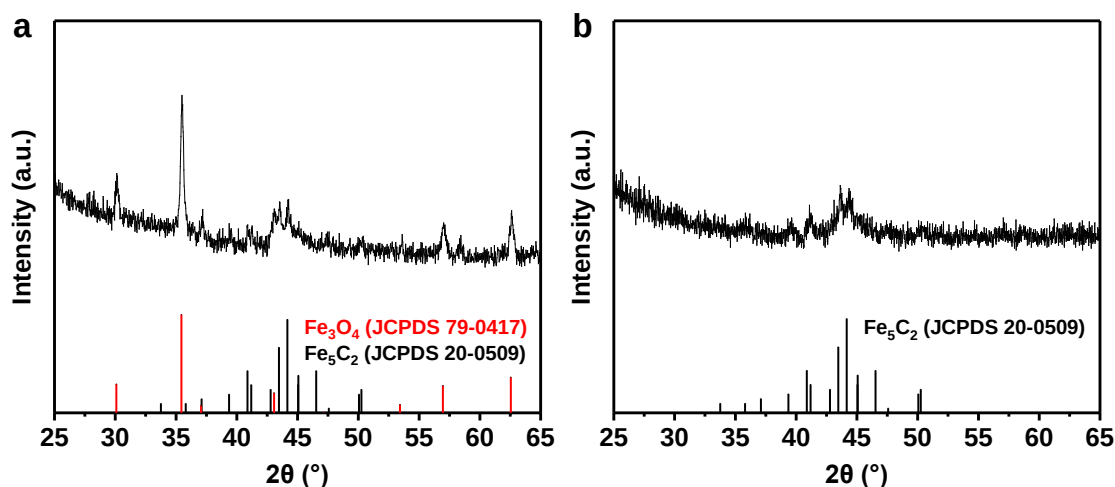


Fig. S16.

XRD patterns of (a) Fe@Si and (b) Fe@Si-c after the test in Fig. 4A. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

Note: In Fig. 4A, the CO₂ selectivity-CO conversion curve was obtained by adjusting the flow rate at a constant H₂/CO ratio. Generally, the traditional Fe-based catalysts were converted into a mixture of iron carbides and oxides during reaction due to the carburization by CO and oxidation by H₂O (13, 24, 25, 27). The carbides oxidation could be easily achieved in the initial stage after exposure of syngas under conventional reaction conditions. In order to obtain conveniently the CO₂ selectivity-CO conversion curve, we checked firstly the relationship between the CO₂ selectivity and CO conversion at conventional reaction conditions (usually at a higher CO conversion), and then analyzed the CO₂ selectivity-CO conversion curve by adjusting the flow rate at a constant H₂/CO ratio. Thus, the data of Fe@Si at CO conversion of ~5% was collected continuously after collecting the data at higher CO conversion levels by adjusting the flow rate. Due to the coexistence of iron carbides and Fe₃O₄ over the Fe@Si catalyst at a higher CO conversion, both iron carbides and Fe₃O₄ still coexisted when collecting the data at CO conversion of ~5% by adjusting the flow rate (as shown in Fig. S16a). Thus, the produced secondary CO₂ was existed over the Fe@Si catalyst. Besides, a primary CO₂ selectivity of about 10% was observed at very low CO conversion over the Fe@Si catalyst, which might be attributed to the presence of Fe₃O₄ phase, as shown in Fig. 4A. Because the hydrophobic shell coated on Fe@Si-c could restrain the oxidation of iron carbides to Fe₃O₄, it was easy to analyze the effect of iron carbides on the CO₂ selectivity. Fe@Si-c was only composed of iron carbides after such test process (as shown in Fig. S16b). No formation of CO₂ was observed at a lower CO conversion (less than 5%), suggesting that only iron carbides existed in the Fe@Si-c catalyst are free of primary CO₂.

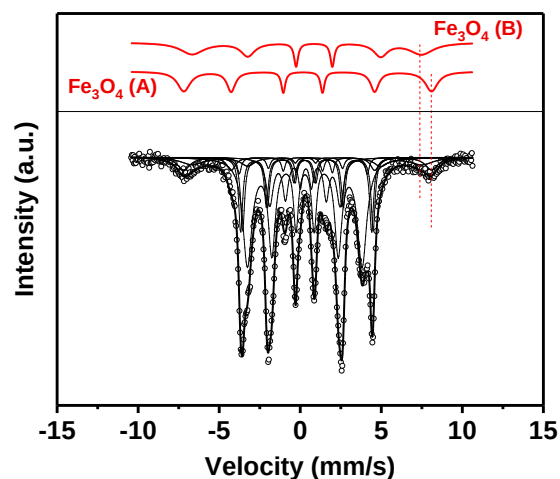


Fig. S17.

Mössbauer spectrum of Fe@Si-c at CO conversion of about 78%. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

Note: At relatively high CO conversion level (about 78%), both iron carbides (85.2%) and little Fe₃O₄ (14.8%) were detected in the Mössbauer spectrum of the spent Fe@Si-c catalyst (Fig. S17 and Table S5), suggesting that oxidation of iron carbides began to occur at much higher CO conversion. At a higher CO conversion level, the amount of water formed would be high, which resulted inevitably in partial oxidation of iron carbides into magnetite before leaving the hydrophobic layer. The formation of Fe₃O₄ enhanced the WGS reaction and promoted the formation of by-produced CO₂. As a result, the CO₂ selectivity reached over 40% at CO conversion of about 78% (Fig. S18).

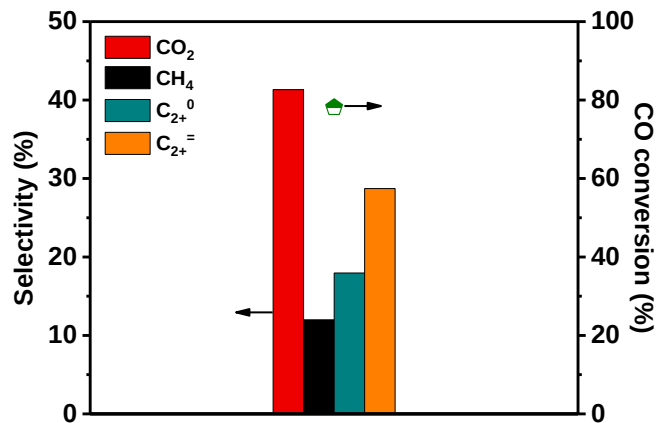


Fig. S18.

Product distribution of Fe@Si-c at relatively high CO conversion. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

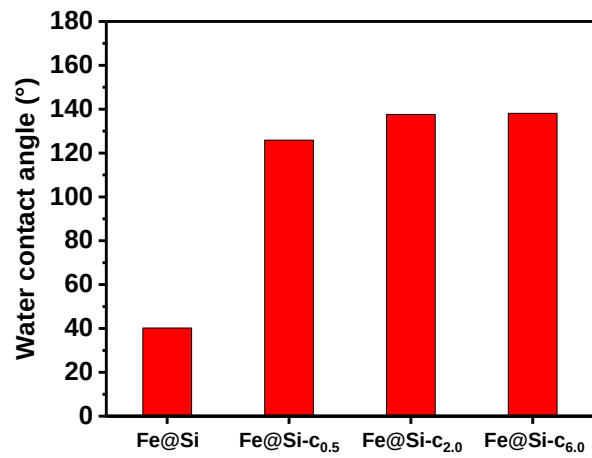


Fig. S19.

Water contact angle over Fe@Si-c with different exposures of TMCS.

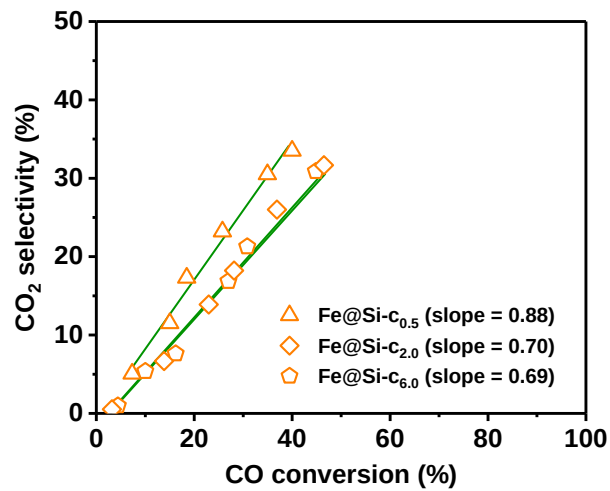


Fig. S20.

CO₂ selectivity-CO conversion curve over Fe@Si-c with different exposures of TMCS.
Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

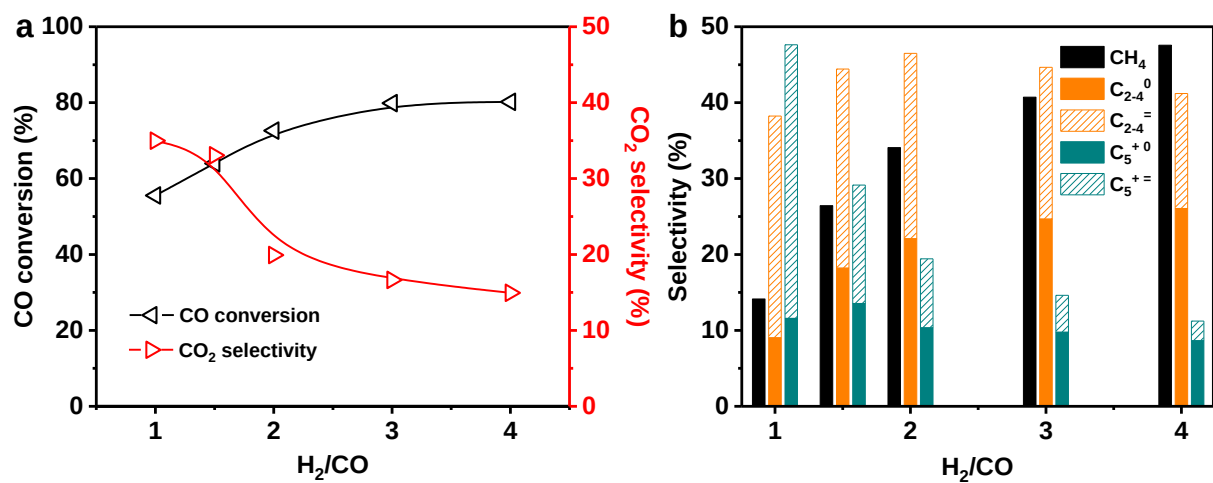


Fig. S21.

Effect of H_2/CO ratio in syngas on (a) CO conversion and CO_2 selectivity and (b) hydrocarbons distribution of Fe@Si-c. Reaction conditions: 320 °C, 2.0 MPa, 4000 ml $h^{-1} g^{-1}$.

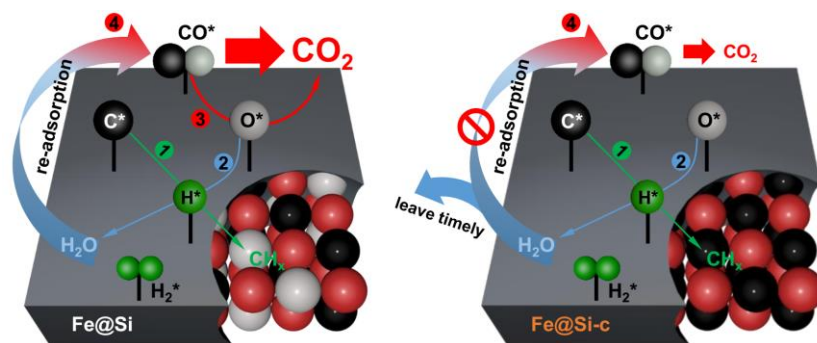


Fig. S22.

Plausible mechanism of hydrophobic modification inhibiting the CO₂ formation.

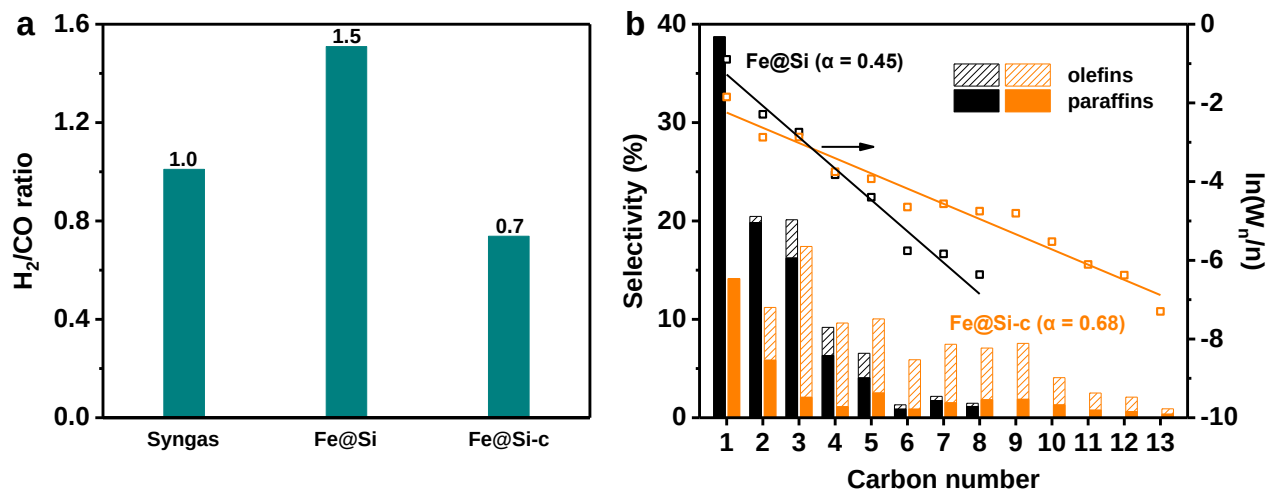


Fig. S23.

Comparison of (a) H_2/CO ratio at the gas outlet and (b) hydrocarbons distribution obtained over Fe@Si and Fe@Si-c. Reaction conditions: 320 °C, 2.0 MPa, $H_2/CO = 1$.

Note: The H_2/CO ratio at the gas outlet in Fig. S23a referred to the mole ratio of H_2 and CO at the gas outlet. And the amount of H_2 and CO at the outlet were determined online by the gas chromatograph. As shown in Fig. S23b, Fe@Si-c exhibited higher chain growth probability and olefins selectivity than Fe@Si. Due to the inhibition of WGS reaction via hydrophobic treatment on Fe@Si, the H_2 amount produced by the WGS reaction decreased, suppressing the hydrogenation and chain termination reactions, thereby enhancing the chain growth probability and olefins production.

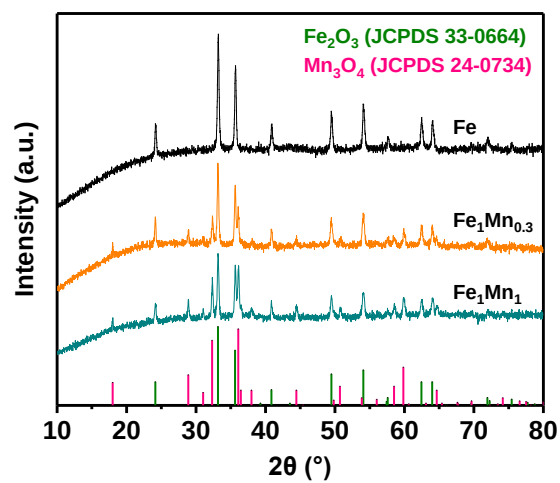


Fig. S24.

XRD patterns of the Fe_1Mn_z catalysts with different Mn/Fe ratios.

Note: As shown in Fig. S24, the added Mn was presented in the form of Mn_3O_4 phase, and no diffraction peak related to Fe-Mn solid solution was detected.

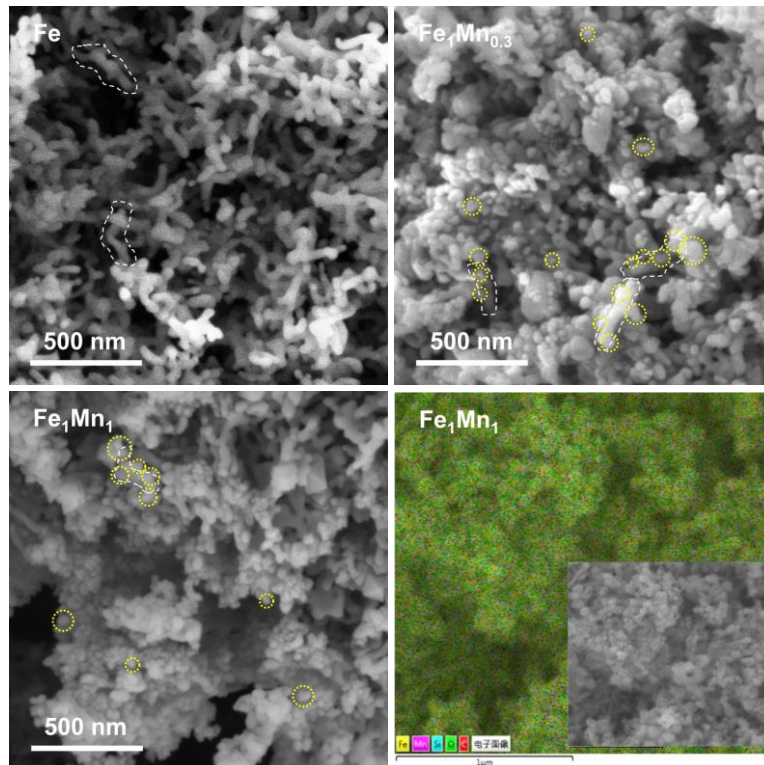


Fig. S25.

SEM images of the Fe_1Mn_z catalysts with different Mn/Fe ratios and relevant EDS elemental mapping. Scale bar in EDS: 1 μm .

Note: As shown in Fig. S25, the single Fe_2O_3 was presented in the form of nanorod-shape (marked by the white dotted line). After the addition of Mn promoter, many microspheres (marked by the golden circle) homogeneously located in the voids between the aggregated Fe_2O_3 nanorods, suggesting that the added Mn was presented in the form of microspheres.

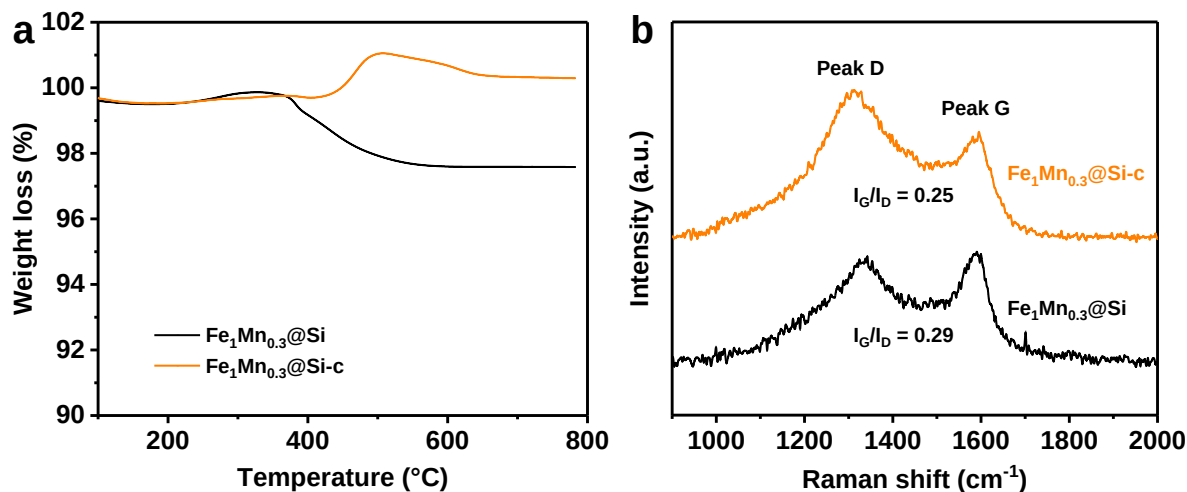


Fig. S26.

(a) TG curves and (b) Raman spectra of the spent $\text{Fe}_1\text{Mn}_{0.3}\text{@Si}$ and $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$.

Note: As shown in Fig. S26a, the coke contents of the two spent catalysts were less than 2.3%, implying that little coke was deposited on the two catalysts. The weight-gain region of the spent $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$ might be attributed to the oxidation process of iron carbides to iron oxides in air. As shown in Fig. S26b, two peaks appeared at 1330 cm^{-1} and 1600 cm^{-1} , which were attributed to the disordered (D type) and ordered (G type) carbonaceous species, respectively (39). $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$ exhibited a lower I_G/I_D ratio than $\text{Fe}_1\text{Mn}_{0.3}\text{@Si}$, indicating more iron carbides active sites. Besides, the carbon balance calculated in each test was between 96% and 98%. As discussed above, the hydrophobic modification could decrease the water content in the vicinity of catalysts and reduce the H_2 amount produced by the WGS reaction. TG and Raman spectra results suggested that the concentration reduction of H_2O and H_2 on the hydrophobic catalysts did not affect obviously the coke deposition and carbon balance during reaction.

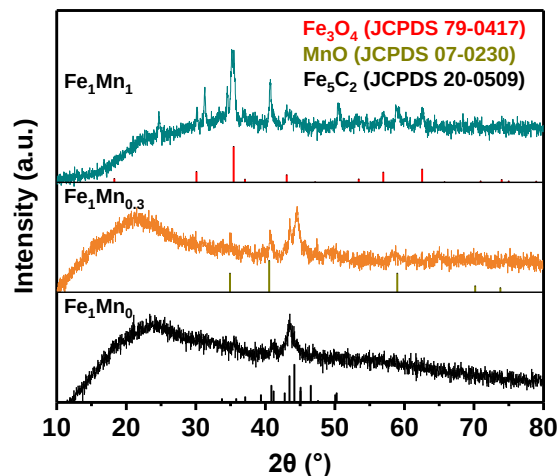


Fig. S27.

XRD patterns of the spent $\text{Fe}_1\text{Mn}_z@\text{Si-c}$ with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, $\text{H}_2/\text{CO} = 2$.

Note: As shown in Fig. S27, only iron carbides were detected in the XRD patterns of the spent $\text{Fe}_1\text{Mn}_z@\text{Si-c}$ catalysts with Mn/Fe ratio ≤ 0.3 . However, both Fe_3O_4 and iron carbides were detected in the XRD pattern of the spent $\text{Fe}_1\text{Mn}_1@\text{Si-c}$ catalyst. This was attributed to that moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO_2 shell, while excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core (Fig. S29). The insufficient shell on the highly promoted catalysts could only inhibit partially the adsorption of water, resulting in partial oxidation of iron carbides into Fe_3O_4 .

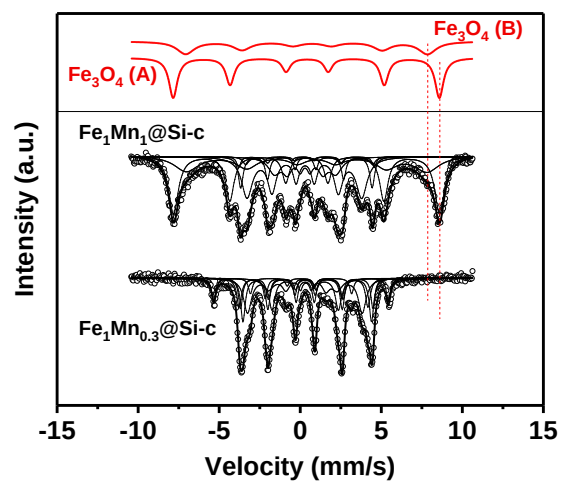


Fig. S28.

Mössbauer spectra of the spent $\text{Fe}_1\text{Mn}_z\text{@Si-c}$ with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, $\text{H}_2/\text{CO} = 2$.

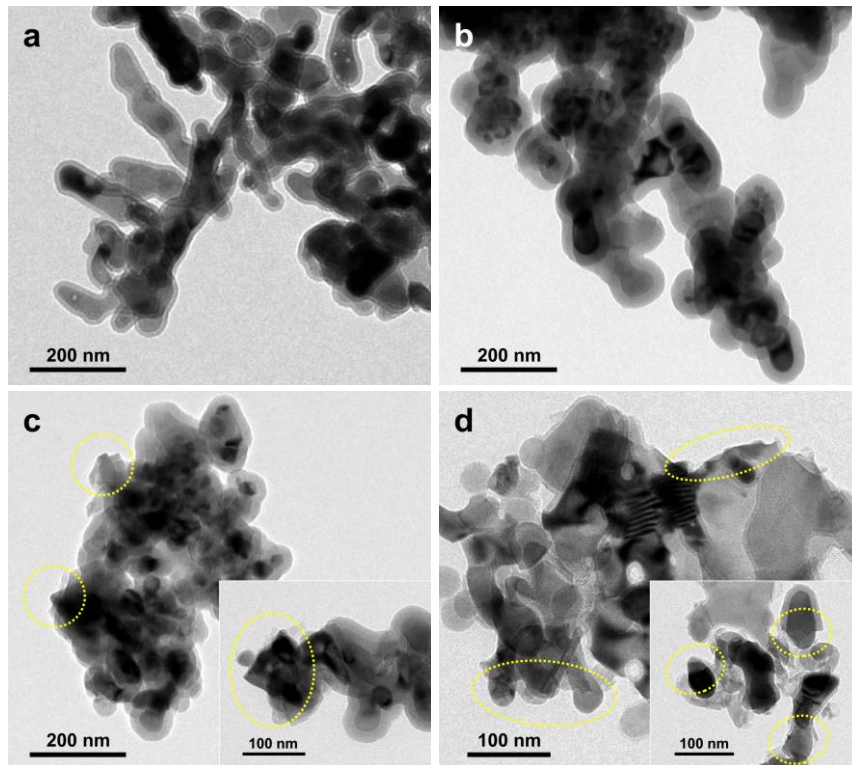


Fig. S29.

TEM images of (a) Fe@Si-c, (b) Fe₁Mn_{0.3}@Si-c, (c) Fe₁Mn_{0.5}@Si-c and (d) Fe₁Mn₁@Si-c.

Note: As shown in Fig. S29, a and b, moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO₂ shell. However, excessive addition of Mn resulted easily in the aggregation of nanorods into large particles (Fig. S29, c and d). Besides, no sufficient shell formation was achieved on the highly promoted catalysts (marked by the golden circle), suggesting that excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core.

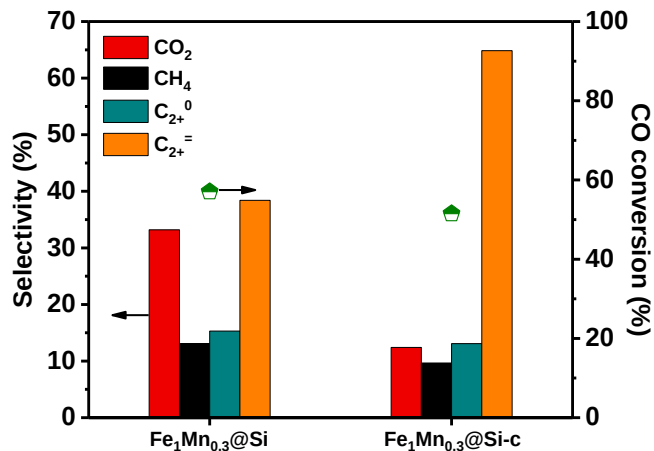


Fig. S30.

Comparison of the product distribution (including CO₂) obtained over Fe₁Mn_{0.3}@Si and Fe₁Mn_{0.3}@Si-c. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

Note: As shown in Fig. S30, the CO₂ selectivity of Fe₁Mn_{0.3}@Si-c was obviously lower than that of Fe₁Mn_{0.3}@Si at a similar CO conversion level, suggesting that the WGS activity was significantly inhibited via the hydrophobic treatment. In addition, compared with Fe₁Mn_{0.3}@Si, Fe₁Mn_{0.3}@Si-c exhibited a suppressed CH₄ selectivity and enhanced olefins selectivity, which was a result of that the H₂ amount produced by the WGS reaction decreased, thereby suppressing the overhydrogenation and chain termination reactions. These results were consistent with the comparison results of Fe@Si and Fe@Si-c mentioned above.

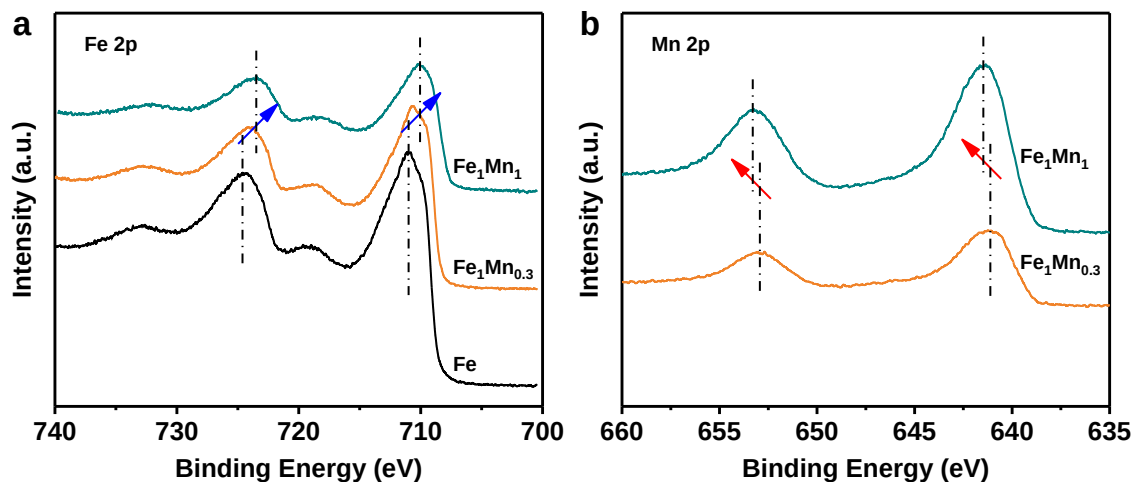


Fig. S31.

XPS spectra of (a) Fe 2p and (b) Mn 2p on the Fe₁Mn_z catalysts with different Mn/Fe ratios.

Note: As shown in Fig. S31, as the increase of Mn content, the Fe 2p peaks gradually shifted to a lower binding energy, while the Mn 2p peaks shifted toward a higher binding energy. These results were attributed to the electron transfer from Mn to Fe atoms (13, 14). This process facilitated the dissociated adsorption of CO and weakened hydrogenation, in favor of C-C coupling reaction and olefins production (14).

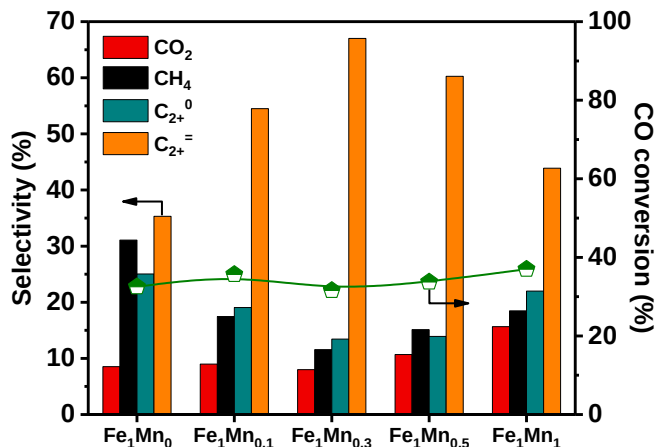


Fig. S32.

Catalytic performance of Fe₁Mn_z@Si-c with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

Note: As shown in Fig. S32, under similar CO conversion level, moderate addition of Mn (Mn/Fe ratio ≤ 0.3) had little influence on the formation of CO₂. Whereas, excessive addition of Mn resulted in the increase of CO₂ selectivity from ~8 to ~16%. This was attributed to that moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO₂ shell, while excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core (Fig. S29). The insufficient shell on the highly promoted catalysts could only inhibit partially the adsorption of water, resulting in partial oxidation of iron carbides into Fe₃O₄ (Figs. S27 and S28) and the enhancement of WGS reaction. Thus, excessive addition of Mn resulted in the increase of CO₂ selectivity.

More importantly, moderate addition of Mn promoted the formation of olefins and suppressed that of CH₄. This was attributed to the enhanced electron transfer from Mn to Fe atoms (Fig. S31), which facilitated the dissociated adsorption of CO and weakened the hydrogenation ability, favoring the C-C coupling reaction and olefins production (14). While, excessive addition of Mn resulted in the enhancement of WGS reaction, which increased the H₂ amount produced by the WGS reaction, thereby facilitating the hydrogenation reaction and suppressing the olefins production.

Table S1.

Comparison of catalytic performance with the previous works.

Catalyst	T ($^{\circ}\text{C}$)	P (MPa)	GHSV (h^{-1})	CO conv. (%)	CO ₂ sel. (%)	Yield of hydrocarbons (%)	Olefins sel. ^d (%)	Yield of olefins (%)	Ref.
ZnCrO _x /SAPO	400	2.5	6828	17	~45	~9.35	80 ^a	~7.48 ^a	(5)
MnO _x /SAPO	400	2.5	4800	8.5	41	5.02	79.2 ^a	3.98 ^a	(6)
ZnZrO _x /SAPO	400	1.0	3600	9.5	45	5.23	63 ^a	3.29 ^a	(7)
CoMn	250	0.1	2000	31.8	47.3	16.8	60.8 ^a	10.2 ^a	(8)
Fe/ α -Al ₂ O ₃	340	2.0	1500	80	40	48.0	53 ^a	25.4 ^a	(9)
Fe-K/NCNTs	300	0.1	4200	16.5	23.6	12.6	54.6 ^a	6.88 ^a	(10)
Mn/Fe ₃ O ₄	320	1.0	4480	41.5	37.8	25.8	60.1 ^a	15.5 ^a	(13)
Fe ₃ O ₄ @MnO ₂	280	2.0	3000	67.9	47.1	35.9	79.3 ^b	28.5 ^b	(14)
KFe@C	340	2.0	30000	~82	47.4	~43.1	44.6 ^a	~19.2 ^a	(15)
Fe/hNCNC	350	0.1	12000	3.5	39.4	2.12	54.1 ^a	1.15 ^a	(16)
FeMn@Si-c	320	3.0	4000	56.1	13.0	48.8	75 ^b	36.6 ^b	TW ^c

^a Light olefins; ^b Total olefins; ^c This work; ^d Olefins selectivity in hydrocarbons.

Table S2.

Detailed Mössbauer parameters (corresponding to Fig. 3, B and C).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	Fe ₃ O ₄ (A)	0.39	0.02	508	33.8
	Fe ₃ O ₄ (B)	0.70	-0.46	469	26.9
	χ -Fe ₅ C ₂ (I)	0.34	0.14	254	5.4
	χ -Fe ₅ C ₂ (II)	0.30	-0.15	220	22.6
	χ -Fe ₅ C ₂ (III)	0.21	-0.16	122	7.7
	θ -Fe ₃ C (a)	0.20	0.30	260	3.6
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.40	0.16	250	18.6
	χ -Fe ₅ C ₂ (II)	0.29	-0.01	220	52.5
	χ -Fe ₅ C ₂ (III)	0.30	-0.08	136	14.7
	θ -Fe ₃ C (a)	0.24	0.20	256	5.7
	θ -Fe ₃ C (b)	0.29	0.05	249	8.5

Table S3.

Detailed Mössbauer parameters (corresponding to Fig. S14).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	χ -Fe ₅ C ₂ (I)	0.33	0.07	250	23.8
	χ -Fe ₅ C ₂ (II)	0.29	-0.02	220	40.8
	χ -Fe ₅ C ₂ (III)	0.25	-0.15	135	17.7
	θ -Fe ₃ C (a)	0.38	0.30	248	11.9
	θ -Fe ₃ C (b)	0.27	-0.19	244	5.8
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.22	-0.07	210	48.8
	χ -Fe ₅ C ₂ (II)	0.20	0.02	190	34.3
	χ -Fe ₅ C ₂ (III)	0.20	0.08	113	16.9

Table S4.

Detailed Mössbauer parameters (corresponding to Fig. S15).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	Fe ₃ O ₄ (A)	0.40	-0.06	510	21.8
	Fe ₃ O ₄ (B)	0.56	-0.34	464	14.1
	χ -Fe ₅ C ₂ (I)	0.35	0.13	254	9.4
	χ -Fe ₅ C ₂ (II)	0.28	-0.06	220	28.9
	χ -Fe ₅ C ₂ (III)	0.30	-0.01	130	13.5
	θ -Fe ₃ C (a)	0.40	0.17	245	4.3
	θ -Fe ₃ C (b)	0.20	0.30	243	8.0
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.23	-0.09	214	37.7
	χ -Fe ₅ C ₂ (II)	0.22	0.06	189	47.8
	χ -Fe ₅ C ₂ (III)	0.21	0.00	115	14.5

Table S5.

Detailed Mössbauer parameters (corresponding to Fig. S17).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si-c	Fe ₃ O ₄ (A)	0.31	0.29	475	7.0
	Fe ₃ O ₄ (B)	0.64	-0.45	440	7.8
	χ -Fe ₅ C ₂ (I)	0.34	0.04	251	12.8
	χ -Fe ₅ C ₂ (II)	0.29	-0.02	220	43.1
	χ -Fe ₅ C ₂ (III)	0.32	-0.06	135	13.9
	θ -Fe ₃ C (a)	0.35	0.24	250	12.6
	θ -Fe ₃ C (b)	0.25	-0.13	245	2.8

Table S6.

Detailed Mössbauer parameters (corresponding to Fig. S28).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe ₁ Mn _{0.3} @Si-c	χ -Fe ₅ C ₂ (I)	0.30	0.19	254	12.9
	χ -Fe ₅ C ₂ (II)	0.30	-0.02	220	34.0
	χ -Fe ₅ C ₂ (III)	0.25	-0.20	132	12.9
	θ -Fe ₃ C (a)	0.36	0.13	246	21.0
	θ -Fe ₃ C (b)	0.27	-0.13	247	14.4
	α -Fe	0.05	/	335	4.8
Fe ₁ Mn ₁ @Si-c	Fe ₃ O ₄ (A)	0.40	-0.02	510	31.3
	Fe ₃ O ₄ (B)	0.70	-0.60	462	21.0
	χ -Fe ₅ C ₂ (I)	0.35	0.05	252	8.2
	χ -Fe ₅ C ₂ (II)	0.28	-0.05	220	24.0
	χ -Fe ₅ C ₂ (III)	0.30	0.00	118	10.6
	θ -Fe ₃ C (a)	0.25	0.25	260	2.4
	θ -Fe ₃ C (b)	0.40	0.30	243	2.5

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